

Qingyang Liu

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Personal Website: <https://xxxco.github.io>

Education

Duke University, School of Medicine, Durham, NC

Aug. 2024 – May 2026

Master of Biostatistics

University of Washington, Seattle, WA

Sept. 2020 – June 2024

BS in Applied Computational Mathematical Science

Experience

Data Scientist II, University of Florida School of Medicine

July. 2026 – Present

- Built multi-stage real-world data pipelines in SQL/R/Python over millions of patient-level records across 5 programs
- Designed RWD validation studies with sample sizing, proportion tests, and non-response weighting for demographics
- Authored methodology documents and SOPs translating technical analyses into deliverables for external stakeholders
- Validated data sources end-to-end with cell-by-cell reconciliation, catching defects missed by industry validators

Machine Learning Research Assistant, Duke University Rohit Singh Lab

Oct. 2024 – May 2026

- Designed a topology-based network selection algorithm using degree scoring across 33 biological networks
- Built DE re-ranking pipeline integrating scRNA-seq signals with network rankings, improving precision @ 10 by 100%
- Reduced pipeline runtime by 4x enabling scalable rare disease gene prioritization at research scale
- Launched a Streamlit ML app for internal use enabling real-time inference and decision-support outputs

Data Science Research Assistant, Duke Center for AIDS Research

Oct. 2024 – May 2026

- Built R pipeline evaluating HIV transmission clustering across 7 genomic subregion and 90+ parameters sets
- Benchmarked ClusterPicker/HIV-TRACE across distance and bootstrap thresholds on 584 sequences
- Identified concordance and clustering yield as distinct performance dimensions via V-measure analysis

Algorithm Engineering Intern, Ping'an Technology

July 2023 – Sept. 2023

- Applied interpolation and curve fitting to improve synchronization between speech and lip movements in AI figures
- Automated analysis of 80,000+ videos in Python, extracting metadata and streamlining workflows for efficiency
- Built and managed databases by uploading and classifying 11,000+ minutes of video data for AI training pipelines
- Processed and optimized raw video data with FFmpeg, adjusting parameters to maximize clarity and visual quality

Projects

Pilot Trial Designing Software, Duke University

Aug. 2024 – Mar. 2026

- Built simulation and Bayesian models to assess pilot-trial feasibility, dropout risk, and data-driven design decisions
- Developed a Shiny app for automated Bayesian scenario testing across sample sizes and prior assumptions

LLM Safety & Interpretability with Sparse Autoencoders, Duke University

Nov. 2025 – Dec. 2025

- Identified toxicity- and hallucination-aligned sparse features from Gemma-2B using a pretrained SAE on 3 datasets
- Trained linear classifiers on top-10 SAE features, achieving 0.96 AUROC on toxicity detection across held-out set
- Applied feature-level activation steering to suppress harmful directions, improving accuracy by 24% on NQOpen

Skills

Programming Language: Python, R, SQL, Java, SAS

Machine Learning: PyTorch, Scikit-learn, LightGBM, TensorFlow, XGBoots, Cross-Validation

Data Analysis: A/B Testing, Statistical Modeling, Pandas, NumPy, Matplotlib, Seaborn, Shiny, Data Visualization

Tools & Platforms: Git, Docker, FFmpeg, LaTeX, Linux